

Year 7 · Chapter 10 · Analysing data: answers (teacher copy)

10.01 Interpreting graphs (booklet p. 3)

WE DO examples

- Ex 1: $4\frac{1}{2}$ symbols = 18 books
- Ex 2: $20 - 8 = 12$ more books
- Ex 3: $\frac{1}{2}$, so 40 students
- Ex 4: July (9°C); June is $19 - 11 = 8^{\circ}\text{C}$ colder than March
- Ex 5: $24\% + 28\% = 52\%$; 52% of 5000 = 2600
- Ex 6: 55%. 20 years ago there was little or no streaming; CDs, radio and downloads were much bigger

Set A

- 4 books
- 2 books
- 12
- 8
- 14
- Thursday
- Wednesday
- 2
- $12 + 18 = 30$
- 4 symbols

Set B

- 6
- 2
- Swimming
- Fishing
- Beach cricket
- 3
- $8 - 2 = 6$
- $6 + 2 = 8$
- 28
- 1 student

Set C

- Bus
- About 20 (a quarter)
- About 10 (an eighth)
- True (6)
- $80 - 40 = 40$
- January and February, 20°C
- Jan and Feb (20°C), Mar and Dec (19°C) or May and Sep (12°C)
- October
- October to November ($+3^{\circ}\text{C}$)
- April to May (-4°C)

Set D

- 28%
- $18\% + 10\% = 28\%$
- 18% of 5000 = 900
- $7\frac{1}{2}$ symbols
- e.g. retirees move to the coast; young people leave for work or uni
- Streaming and radio ($55\% + 20\%$)
- $20\% = \frac{1}{5}$
- 10% of 400 = 40
- 35% of 400 = 140
- e.g. even more streaming, fewer CDs, less radio

Extension

- $60/360 = \frac{1}{6}$; $\frac{1}{6}$ of 240 = 40
- $45/180 = \frac{1}{4}$; $\frac{1}{4}$ of $360^{\circ} = 90^{\circ}$
- $3\frac{3}{4} \times 8 = 30$ students; $20 \div 8 = 2\frac{1}{2}$ symbols

Exit ticket

- Level 1: 6
- Level 2: $9/25$
- Level 3: $5/25 = 20\%$; 20% of 200 = 40

10.02 Misleading graphs (booklet p. 7)

WE DO examples

- Ex 1: the vertical axis starts at 46, not 0, so Ava's lead looks huge
- Ex 2: $52 - 48 = 4$ more votes (not 3 times as many)
- Ex 3: Graph P, because its axis starts at 115 so the bars' heights show only the part above 115
- Ex 4: No. 130 is only 10 more than 120
- Ex 5: column graph starting at 0 with even scale, title and labels
- Ex 6: Ava or her supporters, to make her win look bigger than it is

Set A

- F
- M
- M
- F
- M
- F
- M
- M
- F
- M

Set B

- \$41 000
- 2020
- 39
- Yes
- \$6000 (6 thousand)
- About 7 times
- False
- 2020 to 2021 (+3)
- Yes
- The vertical axis starts at 39, not 0

Set C

- 115
- 0
- 10
- 3 times
- Graph P, because the increase looks much bigger
- 110%
- The percentages add to more than 100%
- 10 years
- 2 years
- Equal gaps show different lengths of time, so the growth looks like it slows down (it is really steady)

Set D

1. Brand C looks about 3 times Brand B
2. B 200 000, C 210 000; difference 10 000
3. Fair column graph from 0
4. Fair line graph from 0
5. The rise looks small and steady, not dramatic
6. No: 54 is not double 52. The rise is only 2
7. It looks 4 times bigger, not 2 times
8. e.g. to sell a product, win a vote, make results look better or worse
9. e.g. title, labelled axes with units, scale starting at 0 with even gaps
10. The missing months could show sales going down, so the trend looks better than it is

Extension

1. 9 times
2. $1.5 \times 1.5 = 2.25$ times bigger; draw it the same size and use $1\frac{1}{2}$ pictures, or make only one direction bigger
3. Misleading: axis starting near 19; fair: axis from 0

Exit ticket

1. Level 1: 80
2. Level 2: 4 points
3. Level 3: the axis starts at 80, so the bars exaggerate the difference; start the axis at 0

10.03 Dot plots (booklet p. 13)

WE DO examples

1. Ex 1: 5 students have 2 pets; 20 students surveyed
2. Ex 2: mode 1; outlier 7
3. Ex 3: dots at 12 (3), 13 (1), 14 (2), 15 (1), 20 (1)
4. Ex 4: cluster from 12 to 15; outlier 20
5. Ex 5: scale 0 to 10; dots at 2 (1), 5 (1), 6 (3), 7 (6), 8 (5), 9 (3), 10 (1)
6. Ex 6: most scores cluster from 6 to 9 (mode 7); 2 is an outlier, so one student may need help

Set A

1. 20
2. 6
3. 4
4. 8 hours
5. 6 hours
6. 12 hours
7. 12
8. 10
9. 4
10. 3

Set B

1. Frequencies: size 6: 2, 7: 4, 8: 6, 9: 3, 10: 1
2. Dot plot matching the table
3. 8
4. 3
5. 4
6. 6
7. 6
8. No
9. $6/16 = 3/8$
10. 10

Set C

1. 26°C
2. 34°C
3. 6
4. 5
5. $3/30 = 1/10$
6. $6/30 = 20\%$
7. 10
8. $7/30$
9. About 24°C to 27°C (most days 23°C to 29°C)
10. e.g. a heatwave or hot westerly wind on one day

Set D

1. Scale 0 to 30 (or 3 to 30); dots: 3 (1), 4 (2), 5 (7), 6 (6), 7 (4), 8 (2), 9 (1), 12 (1), 30 (1)
2. 5
3. 30 (12 is also well away from the cluster)
4. 5
5. $10/25 = 2/5$
6. $6/25 = 24\%$
7. Most values cluster from 4 to 8
8. No. The mode is still 5 because it has the most dots
9. Small data set of whole numbers, so every value can be shown
10. e.g. Most students sent 4 to 8 texts, 5 was most common, and one student sent 30

Extension

1. Any 10 values with the most at 3 and one far away
2. 10 sevens, so 5 eights
3. Too many values and heights have decimals, so the dots would not fit; use a stem-and-leaf plot or grouped graph

Exit ticket

1. Level 1: 4
2. Level 2: $4/12 = 1/3$
3. Level 3: outlier 9; the other values cluster from 2 to 5 (mostly 3 to 4)

10.04 Stem-and-leaf plots (booklet p. 17)

WE DO examples

1. Ex 1: 20, 23, 23, 27
2. Ex 2: 10 people; youngest 12, oldest 46
3. Ex 3: 6 | 2 5 8, 7 | 0 1 3 9, 8 | 0 4 4
4. Ex 4: 4 | 3 7 8, 5 | 5 5 6 9, 6 | 1 2 (key 4 | 3 = 43)
5. Ex 5: 13 | 8 8, 14 | 2 7 9, 15 | 1 3 5, 16 | 0 (key 14 | 2 = 142 cm)
6. Ex 6: 2 | 4 4 8, 3 | 1 5 9, 4 | 0 (key 2 | 4 = 2.4)

Set A

1. 14
2. 24
3. 50
4. 43
5. 5
6. 7
7. 3
8. 8
9. 2
10. 46

Set B

1. 4 | 7
2. 6 | 3
3. 10 | 5
4. 0 | 9
5. 1 2 6 8
6. 0 3 3 4 9
7. 2 | 2 3 6 7, 3 | 1 1 5 8, 4 | 0 4
8. 4
9. 31
10. 44

Set C

1. 20 players; ordered: 7 | 1 5 6 8 9; 8 | 0 0 2 3 4 5 8; 9 | 0 2 2 2 4 8; 10 | 4 5
2. 71 kg
3. 105 kg
4. 92 kg
5. 5
6. 8
7. 7/20
8. $2/20 = 10\%$
9. You can see the lowest, highest and most common values straight away

Set D

1. 13 | 5 8 8, 14 | 2 2 2 6 9, 15 | 1 4, 16 | 0 (key $13 | 5 = 13.5$ s)
2. 11
3. 13.5 s
4. 14.2 s
5. 3
6. 3/11
7. 10 | 9, 11 | 2 8, 12 | 4 4 7, 13 | 1 5, 14 | 0 (key $12 | 4 = 124$)
8. 10 to 14: the first two digits, so every leaf is one digit
9. So the reader knows what the numbers mean (e.g. 2 | 4 could be 24 or 2.4)
10. It shows every actual value, and still shows the shape

Extension

1. 2 | 0 1 3, 2* | 5 6 8, 3 | 1 2, 3* | 7
2. 0.67
3. When one stem has many leaves, so the data is bunched up

Exit ticket

1. Level 1: 8
2. Level 2: 4 | 0 2 6 6
3. Level 3: 5/8; mode 46

10.05 The mean and mode (booklet p. 23)

WE DO examples

1. Ex 1: $30 \div 5 = 6$
2. Ex 2: 4
3. Ex 3: mean 7.33, mode 9
4. Ex 4: mean 2.33, mode 2 (total $28 \div 12$)
5. Ex 5: $5 \times 8 = 40$; $40 - 30 = 10$
6. Ex 6: mean 17.33, mode 12; the mode (12) is better because 44 is an outlier

Set A

1. 4
2. 7
3. 4
4. 20
5. 5
6. 6
7. 5
8. 15
9. 6
10. 10

Set B

1. 3
2. 4
3. 2
4. 6
5. 10
6. 9
7. no mode
8. 2 and 6
9. red
10. 25

Set C

1. mean 4.71, mode 5
2. mean 4.5, mode 2
3. mean 12.83, mode 15
4. mean 4, mode no mode
5. mean 22, mode 21 and 24
6. mean 2, mode 2
7. mean 20, mode 20
8. mean 2.1, mode 2
9. mean 1.3, mode 1
10. mean 151.67, mode no mode

Set D

1. $24 - 19 = 5$
2. $360 - 293 = 67$
3. e.g. 4, 4, 5, 8, 9
4. mean 12, mode 6; the mode, because 40 is an outlier
5. Mean goes from 7 to 9.17 (up about 2.17)
6. $5 \times 8 = 40$; $40 - 30 = 10$
7. $90 - 70 = 20$
8. The mode (size 8), because it is the size sold most often
9. 16, 17, 18
10. Both means are 16; A has mode 15, B has no mode

Extension

1. $650 + 15 = 665$; $665 \div 10 = 66.5$
2. $96 - 14 = 82$; $82 \div 6 = 13.67$
3. Yes, e.g. 1, 2, 3, 4 has mean 2.5

Exit ticket

1. Level 1: 8
2. Level 2: mean 5.5, mode 7
3. Level 3: $36 - 24 = 12$

10.06 The median and range (booklet p. 27)

WE DO examples

1. Ex 1: 3, 4, 4, 5, 7, 8, 9 $\rightarrow$ 5
2. Ex 2: $20 - 5 = 15$
3. Ex 3: median 7.5, range 9
4. Ex 4: median 35, range 18
5. Ex 5: median 12, range 33, mean 17.33; the median, because 44 is an outlier
6. Ex 6: e.g. 3, 5, 8, 9, 13

Set A

1. 3
2. 4
3. 5
4. 6
5. 11
6. 7
7. 14
8. 3
9. 22
10. 7

Set B

1. 7
2. 17
3. 5
4. 12
5. 15
6. 0
7. 38
8. 4.3
9. 15
10. 7

Set C

1. median 5.5, range 7
2. median 14.5, range 9
3. median 3.1, range 3.6
4. median 7, range 9
5. median 1, range 9
6. median 22, range 5
7. median 2, range 3
8. median 151, range 15
9. median 175, range 200
10. median 1, range 3

Set D

1. e.g. 2, 4, 6, 8, 10
2. $7 + 15 = 22$
3. Median $7 \rightarrow 7.5$; range $6 \rightarrow 16$
4. Class B is more consistent (smaller range); Class A did better overall (higher median)
5. The mean uses every value's size, but the median only depends on the middle position
6. 10 and 10
7. e.g. 3, 6, 8, 10, 13
8. 195 cm
9. Yes, when every value is the same
10. median 12.5, range 41. Not really: no one is aged about 12.5

Extension

1. The 20th and 21st
2. 20 values: 10th and 11th are both 3, so the median is 3
3. 50.5

Exit ticket

1. Level 1: 7
2. Level 2: median 6.5, range 8
3. Level 3: e.g. 2, 4, 6, 9, 12

10.07 Analysing dot plots and stem-and-leaf plots (booklet p. 31)

WE DO examples

1. Ex 1: range 5, mode 1
2. Ex 2: median 1, mean 1.6
3. Ex 3: range 4, median 4.5, mode 6, mean 4.25
4. Ex 4: range 27, median 29, mode 34, mean 28.33
5. Ex 5: with 15: mean 3.45, median 2; without: mean 2.3, median 2
6. Ex 6: the median (or mode), because the outlier 15 pulls the mean up

Set A

1. 15
2. 6
3. 6
4. 2
5. 2
6. 32
7. 2.13
8. 6
9. 5
10. 2/15

Set B

1. 10
2. 24
3. 53
4. 29
5. 31
6. 36.5
7. 367
8. 36.7
9. 2
10. 35 years

Set C

1. 9
2. 2
3. 2.5
4. 2.9
5. 9; it makes the mean bigger
6. 35 min
7. 25 min
8. 25 min
9. 24.86 min
10. 3

Set D

1. range 5, median 8, mode 8, mean 7.65
2. mean 7.38, median 8
3. The mean: the low score of 2 lowers the total, but the middle position barely moves
4. No: total is 153; for a mean of 8 with 21 students the total must be 168, so they would need 15 out of 10
5. range 53, median 21, mode 21, mean 24.47
6. 68: it is far from all the other ages (15 to 34)
7. mean 21.36, range 19
8. The median (or mode), because the outlier pulls the mean up
9. e.g. 2, 4, 4, 4, 5, 6, 7, 8, 8 drawn as a dot plot (mode 4, median 5, range 6)
10. Values are balanced on both sides of the middle, so the mean sits in the middle too

Extension

1. Total $5 \times 28 = 140$; $140 - 112 = 28$, so the leaf is 8
2. Any value of 6 or more
3. e.g. 1, 2, 3, 4, 5, 30: one large value pulls the mean up

Exit ticket

1. Level 1: 3
2. Level 2: median 3, range 7
3. Level 3: mean 2.91; without the 8 it drops to 2.4

10.08 Comparing data sets (booklet p. 37)

WE DO examples

1. Ex 1: A mode 7, range 5; B modes 8 and 9, range 8
2. Ex 2: A median 7, B median 7.5. B did slightly better (higher median); A is more consistent (smaller range)
3. Ex 3: plot with stems 2–5, key e.g. 3 | 4 = 34
4. Ex 4: P median 40, range 24; Q median 32, range 17. P did better (higher median); Q is more consistent (smaller range)
5. Ex 5: Amy mean 13.33, range 3; Beth mean 14, range 17. Amy is more consistent
6. Ex 6: e.g. Amy: similar mean but far more reliable. (Accept Beth if argued: she can score very high.)

Set A

1. 12
2. 15
3. 18
4. 63
5. 25
6. 35
7. 33
8. 42
9. 9
10. Gym B (median 39 vs 32)

Set B

1. P
2. S
3. P
4. S
5. P
6. S
7. P
8. S
9. P
10. S

Set C

1. Stems 14–17; girls' leaves on the left
2. 175 cm, boys
3. girls 160, boys 158
4. girls 161, boys 158
5. girls 22, boys 30
6. girls 159.18, boys 158.5
7. The girls have a slightly higher mean and median, but the squads are very similar
8. The girls (smaller range)
9. girls 6, boys 4
10. e.g. The squads have similar average heights. The boys' heights are more spread out.
11. Primary
12. e.g. the girls: higher mean and median, and more consistent heights

Set D

1. Stems 2–9; 7R leaves on the left
2. 97, 7T
3. mean 67.47, median 68, mode 72, range 41
4. mean 70.67, median 74, mode no mode, range 74
5. 7T has a higher mean (70.67 vs 67.47) and median (74 vs 68)
6. 7R: range 41 vs 74
7. 23: it is well below the rest of 7T
8. mean 74.07, range 59
9. Primary: she collected the marks herself
10. e.g. 7T scored higher on average but their marks were much more spread out; 7R was more consistent

Extension

1. $96 + 204 = 300$; $300 \div 20 = 15$
2. The bigger group has more effect on the total
3. e.g. 5, 5, 5, 5, 5 and 1, 3, 5, 7, 9

Exit ticket

1. Level 1: 30, Team A
2. Level 2: A 22.5, B 26
3. Level 3: Team B (range 11 vs 18)